

Report

Course Project: Turbulence at finite Reynolds number and the skeleton of chaos.

Name: Sofia Sannino

Date: May 21, 2026
Course: Turbulence ME-467
Instructor: Tobias Schneider

Part I: Intermittency corrections to K41

1 Data Analysis

1.1 Velocity Signal in the Spatial Domain

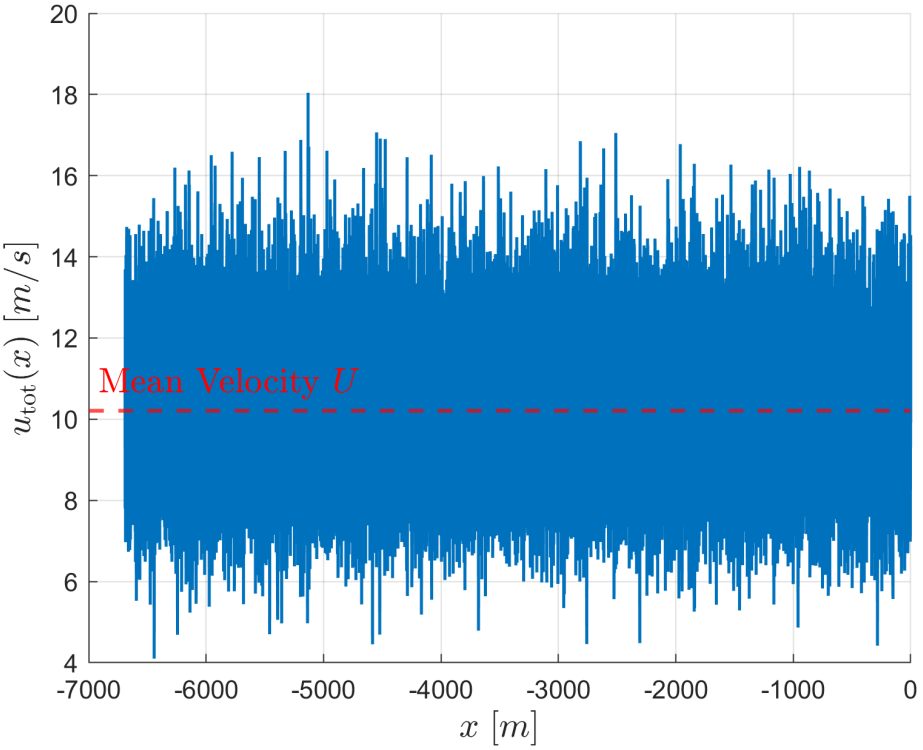


Figure 1: Plot A: using $x = -U/f$ convention for Taylor's Frozen Flow hypothesis, with f sampling frequency.

(Max. 3 lines) To what extent are spatial interpretations of the time series appropriate?
Taylor's frozen hypothesis is justified if $I \ll 1$. In our case $I \sim O(10^{-1})$, thus the spatial interpretation is reasonable but introduces a non-negligible approximation, especially at larger scales, where the characteristic velocity u_l is higher.

1.2 Correlation Length of the Velocity Signal

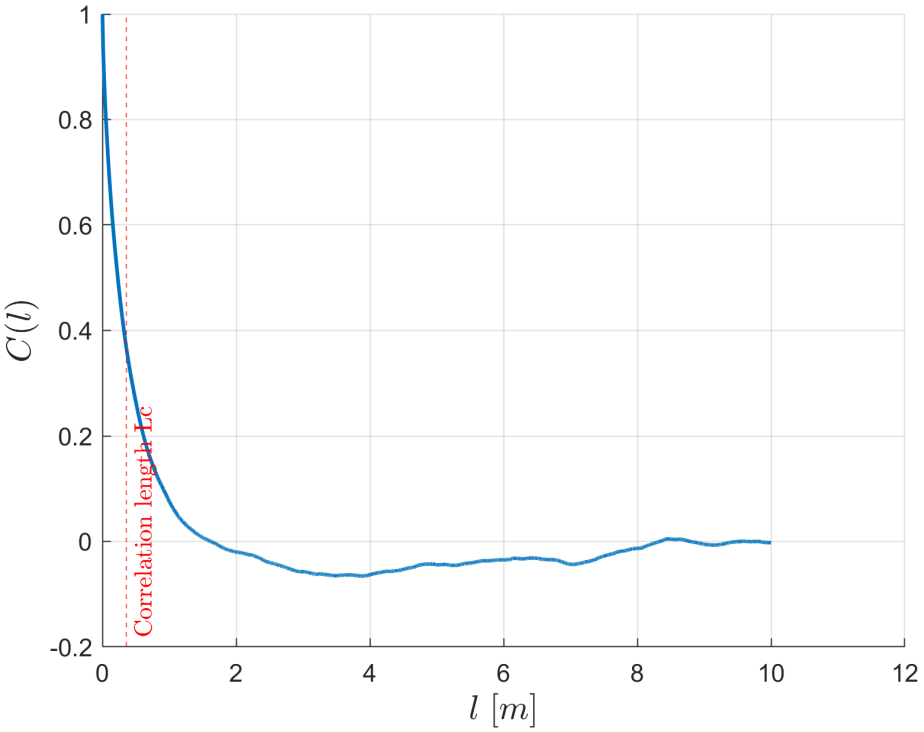


Figure 2: Plot B

1.3 Energy Spectrum of the Flow

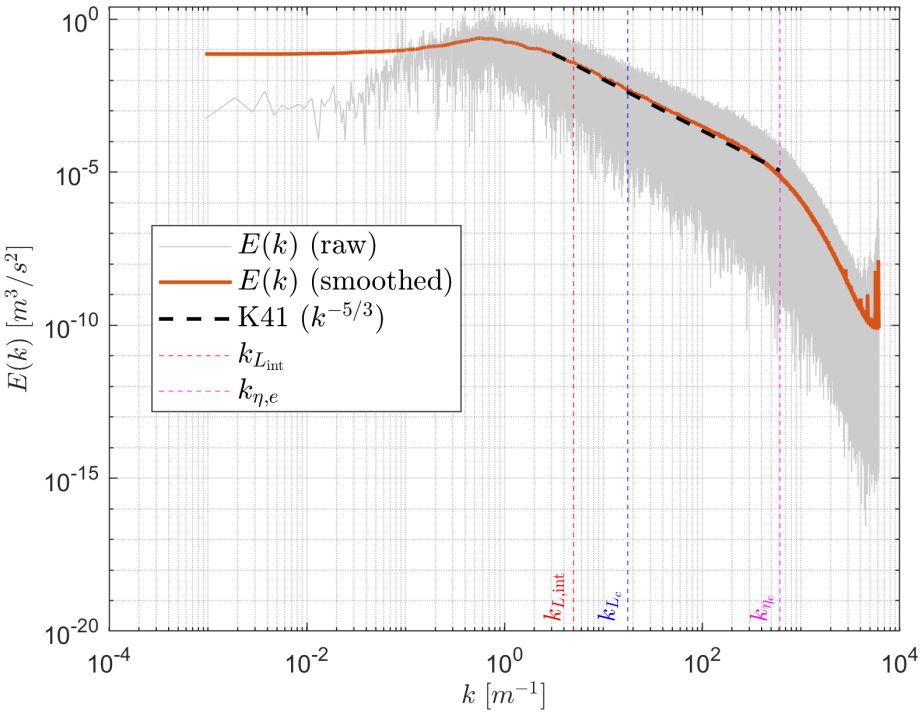


Figure 3: Plot C

(Max. 3 lines) Indicate predictions of the K41 theory on Plot C. Does the data follow the theory predictions?
Consistently with K41, the data approximately follow $E(k) \sim k^{-5/3}$ over an intermediate k range, thus corresponding to the inertial range,from which k_{int} and k_η are estimated.As expected,at low k high,nearly constant, energy values are observed, and at high k the energy drops because of dissipation.

1.4 The Dissipation Rate and Different Reynolds Numbers

See the results in Table 1.

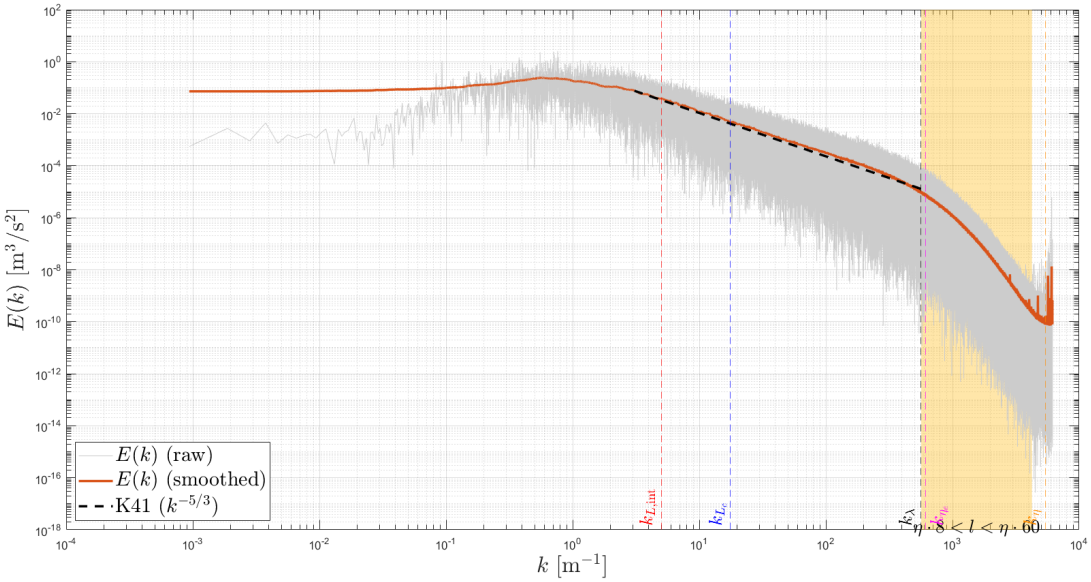


Figure 4: Plot C with wave numbers k_λ (black) and k_η (orange), obtained through formulas, and Pope's dissipation range.

1.5 Velocity Increments

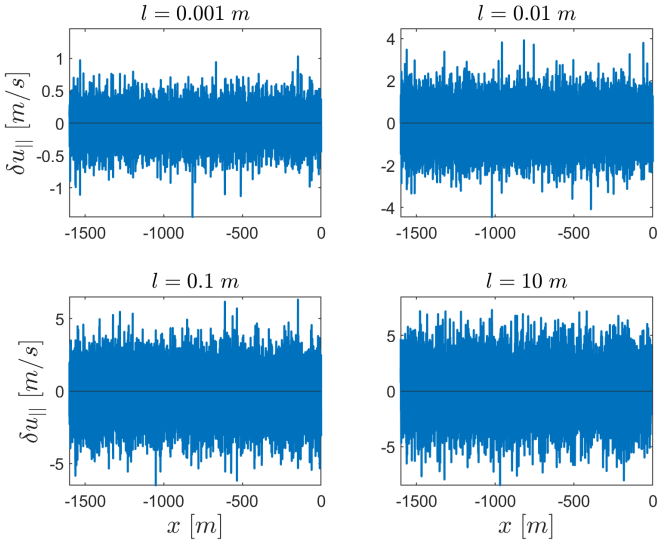


Figure 5: Plot D

(Max. 7 lines) Where do these l-values lie relative to the turbulent length scales you determined from the energy spectrum? Describe the differences in the signal between small and large length scales. Explain whether or not your observations can reveal the trend in the autocorrelation curve you plotted in Section 1.2.
Clockwise, considering for each l the corresponding k, the first increment lies in the dissipation range ($k \sim 6.28 \cdot 10^3$), the second lies in Kolmogorov-scale region ($k \sim 6.28 \cdot 10^2$), the third lies in the inertial range ($k \sim 6.28 \cdot 10^1$) and the last lies in the energy-containing range ($k \sim 6.28 \cdot 10^{-1}$). From 5, one can notice that the amplitude of the longitudinal increments increases with l. This is consistent with the trend observed in 2: low l means high $C(l)$, i.e. that the two signals $u(x+l)$ and $u(x)$ are highly correlated, thus that they will likely have a similar value, giving correspondent small $|\delta u_{\|}(x,l)|$. An analogous logic stands for high l: poorly correlated signals will more likely give larger $|\delta u_{\|}(x,l)|$ values.

1.6 Statistics of Velocity Increments

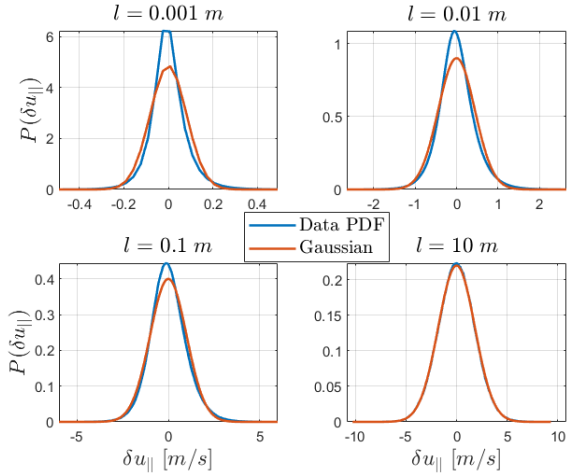


Figure 6: Plot E1

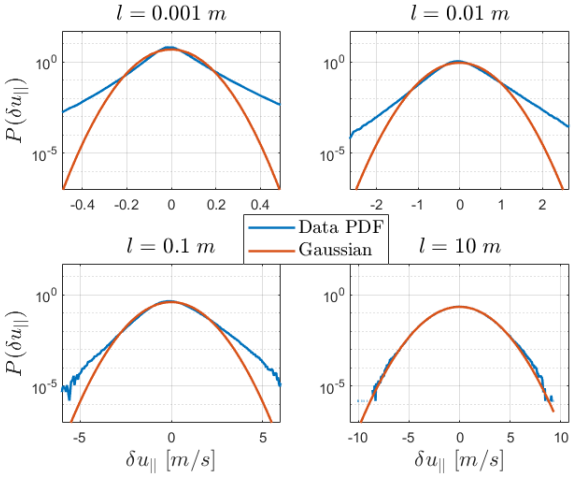


Figure 7: Plot E2

(Max. 5 lines) Discuss how the distribution changes with l. Also discuss in what way the l-dependence of the PDF supports or contradicts the assumption of self-similarity, on which K41 is based.
For $l \ll l_0$, with the Gaussian as a reference, PDFs appear increasingly more peaked and with heavier tails as l decreases, suggesting intermittency and that the distribution is not just rescaled but changes shape. As l increases, the PDF becomes gaussian, as turbulence is driven by larger eddies, with a poorly correlated velocity field. Moreover, confronting consecutive l p-moments,the scaling exponent changes with p with order $\sim O(10^{-1})$. These observations suggest deviations from self-similarity.

1.7 Structure Functions and Energy Dissipation

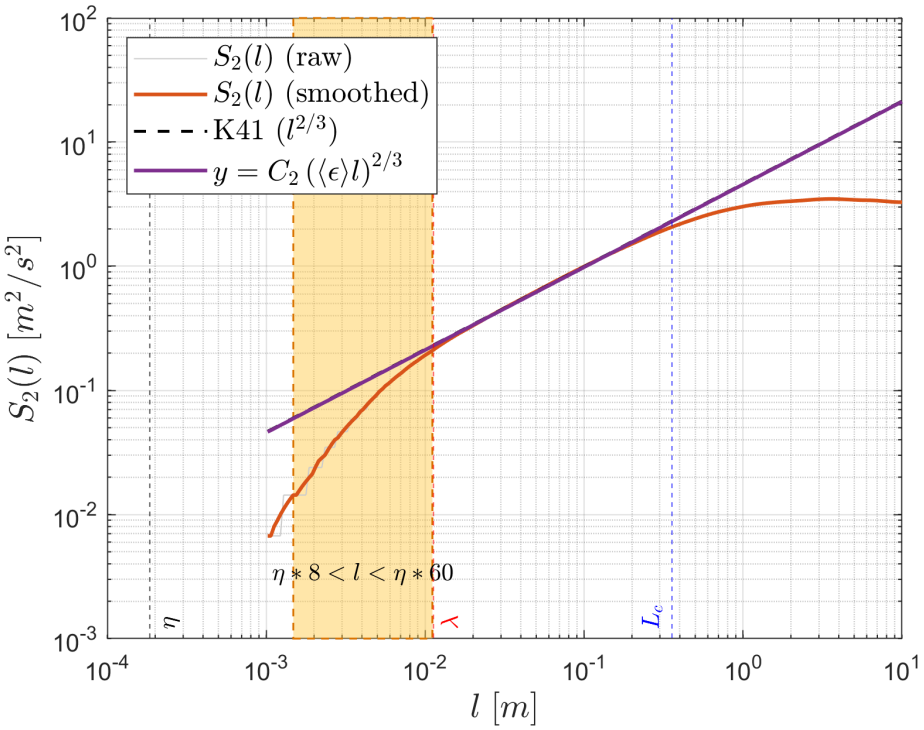


Figure 8: Plot F: Pope's dissipation range included.

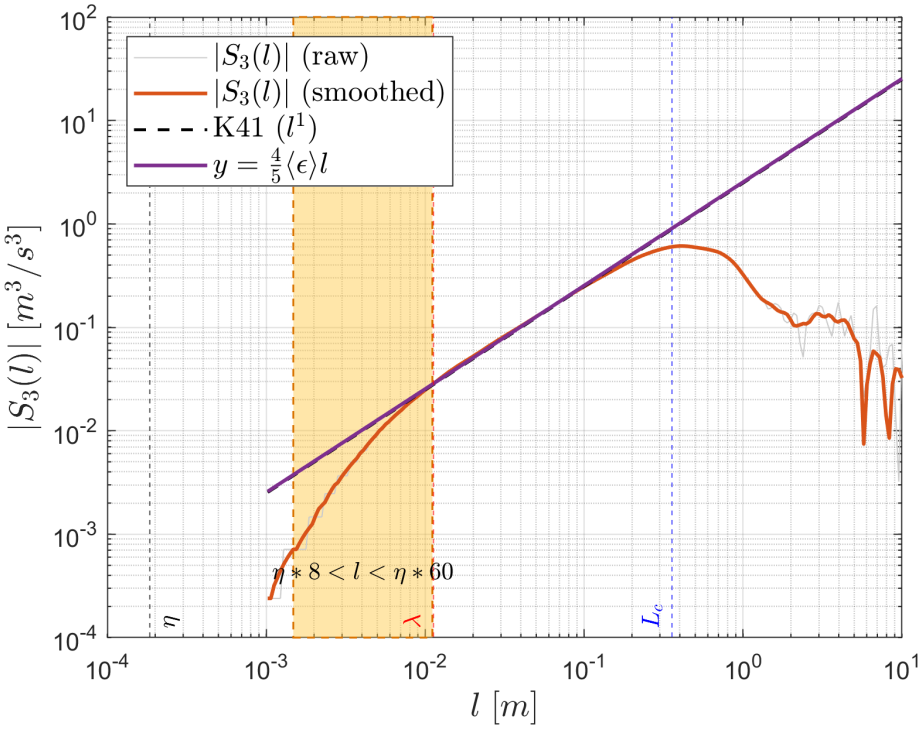


Figure 9: Plot G: Pope's dissipation range included.

(Max. 3 lines) K41 predicts that $S2(l)$ scales as l^{ζ_2}. How is the scaling of $S2(l)$ related to the scaling of the energy spectrum E(k)?
By flow homogeneity, $S_2(l) = \langle [u(x+l) - u(x)]^2 \rangle = 2[\Gamma(0) - \Gamma(l)]$, where Γ is two-point correlation function. Using Wiener-Khinchin and the evenness of $E(k)$, $S_2(l) = 2\int_{\mathbb{R}}[1 - \cos(kl)]E(k)\,dk$. If $E(k) \sim k^{-n}$, then, by the change of variable $q = kl$, $S_2(l) \sim l^{n-1}$, thus $E(k) \sim k^{-(\zeta_2+1)} = k^{-5/3}$.

(Max. 8 lines) [...] Discuss the relevance of [L_c and λ] for the K41 predictions of S_2 and S_3. Also discuss to which extent your data supports the K41 predictions.
The 4/5 law for S_3 follows from the Kármán-Howarth-Monin relation in the inertial-range limit: the scale must be far from the forcing range, $l \ll L_c$, and direct viscous effects must be negligible. Thus, L_c gives the upper bound of the inertial range, while λ gives a conservative lower bound,since without self-similarity (H2) one can only guarantee negligible dissipation for $l \gg \lambda$. Hence $[\lambda, L_c]$ is a minimal inertial range for the S_3 prediction.S_2 law is obtained under the additional assumption H2. With H2,one can show that dissipation is negligible until η and Pope's [2] estimate places dissipative effects around $8\eta < l < 60\eta$,extending the expected inertial range below λ.Data shows that S_2 and S_3 are compatible with K41 inside $[\lambda, L_c]$, while deviations appear as the scales approach Pope's dissipation range.

(Max. 3 lines) How can you estimate the energy dissipation rate ε from $S_2(l)$ and $S_3(l)$? Discuss how the estimates compare to the estimates based on integral-scale quantities in Section 1.4.
Giving $S_2(l) = C_2(\varepsilon l)^{2/3}$ and $S_3(l) = -(4/5)\varepsilon l$, ε could be estimated over (λ, L_c), where they stand, by inverting the formulas.The obtained values are about 5-7% larger than ε_{int},likely due to an approximated inertial range, or numerical errors, suggesting coherence between the ε_{int} formula and S_2/S_3 ones.

1.8 Flatness of Velocity Increments

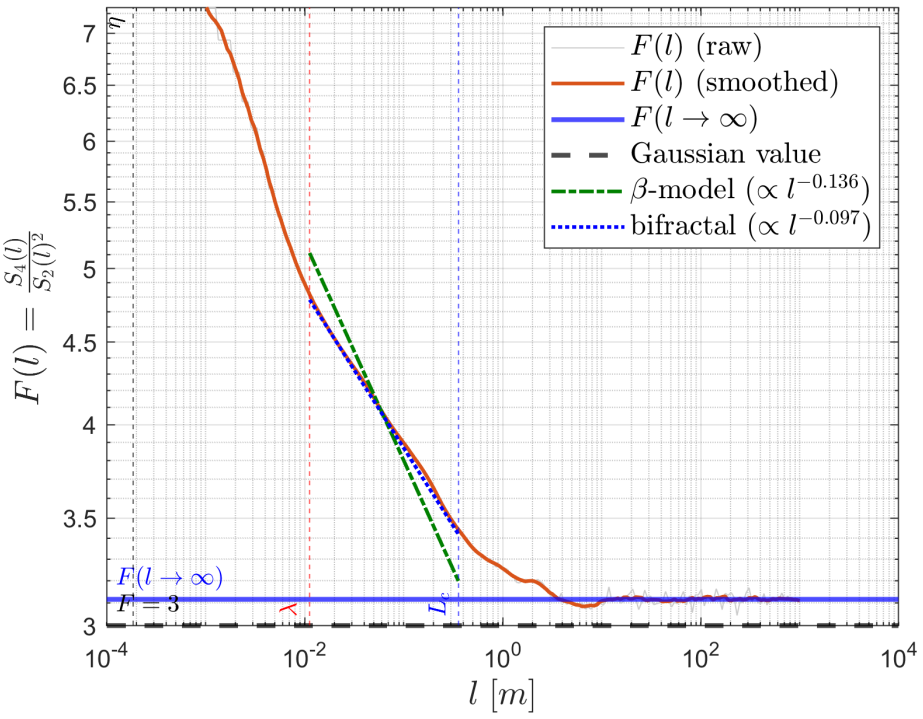


Figure 10: Plot H

(Max. 3 lines) Describe how the flatness variation relates to changes in the shape of the PDFs of $\delta u_{||}$.
According to K41, $F(l)$ would be constant in the inertial range,but it increases as l decreases. As l is

small, $F(l)$ is increasingly greater than $\frac{\langle X^4 \rangle}{\langle X^2 \rangle^2} = 3$, X Gaussian, which means that $\delta u_{||}$ has increasingly heavier tails and higher peaks for small l, as seen in 1.6, diverging from a Gaussian PDF.

(Max. 3 lines) Does $F(l)$ converge to a fixed value for large l? If so, what value does it seem to approach ?
As l goes to energy-containing scales, $F(l)$ converges to approximately 3, the gaussian value. This is consistent with 1.6: at larger scales, $\delta u_{||}$ has a nearly Gaussian distribution, suggesting that turbulent fluctuations are more homogeneously distributed across space, in a poorly correlated velocity field.

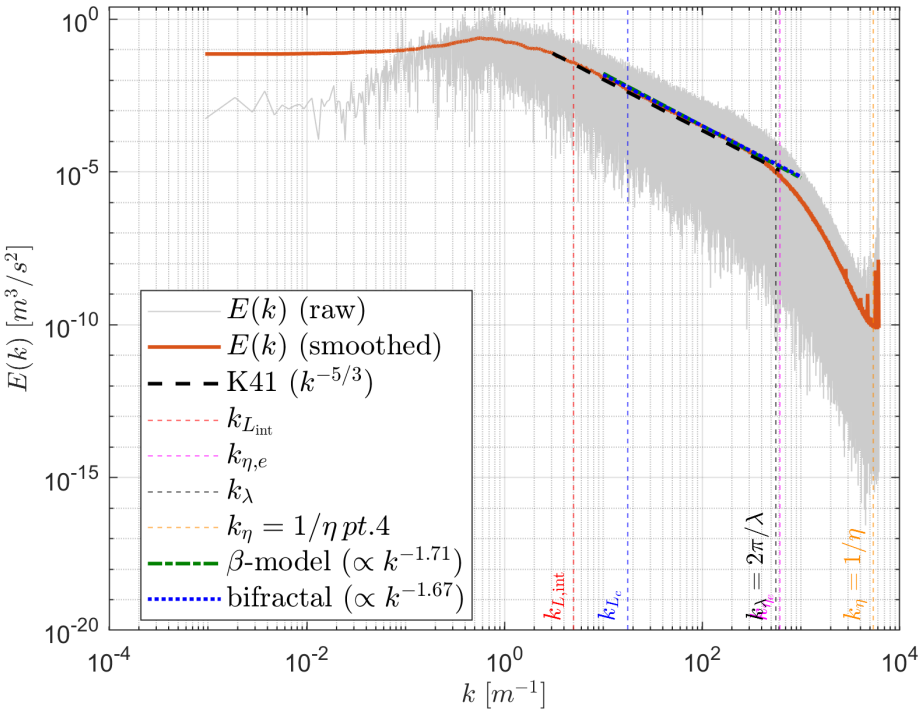


Figure 11: Plot C2: correction to energy spectrum

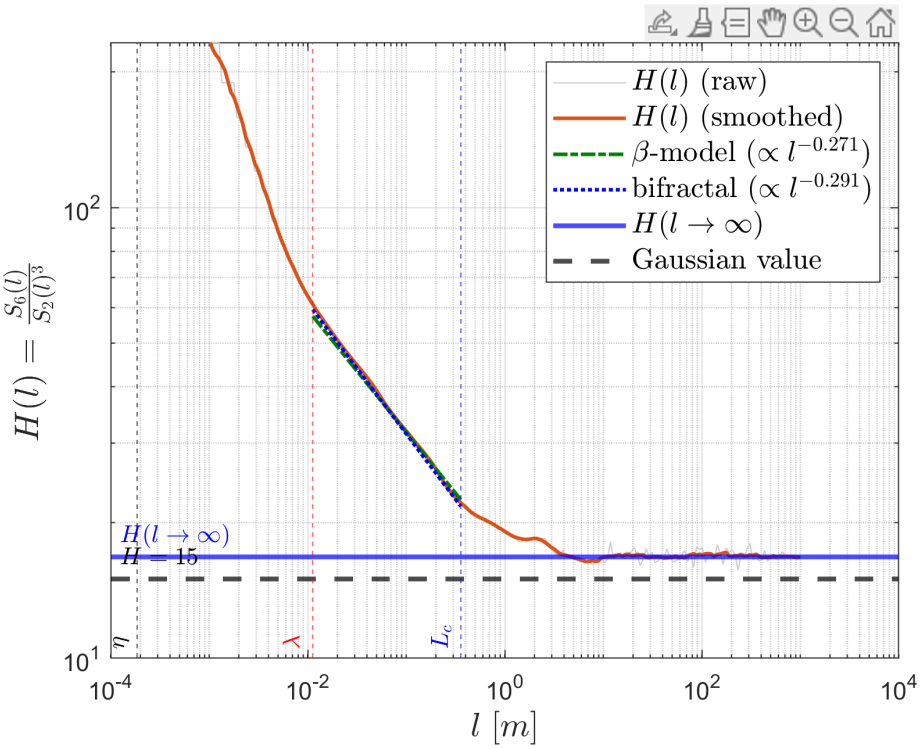


Figure 12: Plot I

2 Interpretation and Discussion

(Max. 16 lines) Based on the data analysis, to which extent does the K41 theory derived for Re $\to \infty$ limit describe the "real" data at finite Reynold's number? Can all the computed quantities of the data analysis be trusted?

This experiment validates K41 partially, framing it as an useful approximation, that has limitations due to intermittency, that breaks self-similarity, and to finite Reynolds number in real world. In 1.6 and 1.8, the observations on $\delta u_{\parallel}$ distribution, the study of the scaling exponent and the pattern of F for small scales tell us that self-similarity, on which the structure functions calculation is based, is not true for $l \ll l_0$ due to intermittency and that dissipation will impact smaller scales sooner than expected by K41. Despite this, in 1.3 and in 1.7 K41 predictions match data : the energy pattern shows large scales at which energy is injected, an inertial range in which the scaling $k^{-5/3}$ is followed, and a dissipation range where energy drops, although the dissipation range should be closer to η if self-similarity was fully respected, and, likewise, S_2 and S_3 follow respectively k and $k^{2/3}$ scaling, again in an inertial range that should be larger if self-similarity held. This considerations tell us that Kolmogorov is a valid and useful model for turbulence in inertial ranges, but it is by construction limited since it assumes an asymptotic behavior that could not match reality, as flows with infinite Reynolds number do not exist and self-similarity is always broken by intermittency in real turbulent phenomena. Given these considerations, K41, completed with intermittency corrections models, remains a powerful framework to simulate and study turbulence.

Table 1: Results table.

#	Quantity	Dimension	Value
1	U	ms^{-1}	10.20864
2	I	-	$1.25448 \cdot 10^{-1}$
3	$C(l)$ normalization	m^2s^{-2}	1.64007
4	L_c	m	$3.57303 \cdot 10^{-1}$
5	L_{int}	m	$3.52871 \cdot 10^{-1}$
6	$\tilde{E}(k)$ normalization (analytical)	m	$\frac{L}{2\pi N^2}$
7	$\tilde{E}(k)$ normalization (numerical)	m	$6.19795 \cdot 10^{-12}$
8	Parsival rel. error	-	$1.21848 \cdot 10^{-15}$
9	$L_{\text{int},e}$	m	1.25664
10	η_e	m	$1.02166 \cdot 10^{-2}$
11	ε from int. scale	m^2s^{-3}	2.93922
12	Re_λ	-	953.46
13	Re	-	30303.15
14	Re_η	-	1
15	ζ_2	-	2/3
16	ζ_3	-	1
17	ε from $S_2(l)$	m^2s^{-3}	3.09157
18	ε from $S_3(l)$	m^2s^{-3}	3.16782
19	$\tilde{E}(k)$ scaling, β-model	-	$E(k) \sim k^{-1.71185}$
20	$\tilde{E}(k)$ scaling, bifrac. model	-	$E(k) \sim k^{-5/3}$
21	$H(l)$ scaling, β-model	-	$H(l) \sim l^{-0.27107}$
22	$H(l)$ scaling, bifrac. model	-	$H(l) \sim l^{-0.29075}$

3 Part II: Chaos, its skeleton, and quantitative characterization

3.1 Chaotic behavior of the KSE

3.1.1 Qualitative investigation

Discussion: $\nu > 0$ is the hyperviscosity parameter that controls KSE dynamics determining the impact of dissipation on it and the stability of perturbed solutions.

- $\nu = 10^{-2}$: In this first case, the perturbed solution and the perturbed one look approximately the same in the $(x, t/\nu)$ space, showing also an almost vanishing separation in the third plot. In the phase portrait, the unperturbed solution is a stable fixed point and the perturbed solution, after a transient, reaches it. Thus, it is a stable orbit.
- $\nu = 10^{-3}$: In this case, again, the perturbed solution looks approximately identical to the unperturbed one in the $(x, t/\nu)$ space, with an almost null separation between the two trajectories. The phase portrait shows that the unperturbed solution is a stable fixed point, while the perturbed one displays a short transient before converging to the same fixed point. Thus, it is a stable orbit.
- $\nu = 8 \cdot 10^{-4}$ In this third case, the two solutions look very similar, but the separation plot shows a regular and periodic pattern, with higher values around $|x| \in [0.2, 0.4]$. The phase portrait shows two nearly overlapping closed loops, suggesting convergence toward a stable periodic orbit (limit cycle) with a small phase shift between the two trajectories. This is validated also by the periodic and regular pattern of separation, with a space pattern fixed and repetitive in the same range.
- $\nu = 6.5 \cdot 10^{-4}$ In this fourth case, the two solutions look, again, very similar in the $(x, t/\nu)$ plot. But the separation pattern shows that separation between the two trajectories increases over time. Indeed, the phase portrait shows the unperturbed solution going along trajectories along two periodic orbits, with the perturbed one that follows the same pattern but with increasing separation from the first, that will probably grow exponentially. Thus, the periodic orbits, giving the pattern of separation, are probably unstable and the trajectories lie in a chaotic attractor.
- $\nu = 2 \cdot 10^{-4}$ In this last case, the solutions in $(x, t/\nu)$ plots are clearly different. The separation pattern suggests they will follow different trajectories, indicating sensitive dependence on initial conditions. Indeed, the phase portrait shows that both the perturbed solution and the unperturbed one remain bounded but follow different paths through several periodic orbits. Thus, again, the orbits are unstable and the trajectories evolve on a chaotic attractor.

Overall, it can be observed that decreasing ν weakens the hyperviscous dissipation and the dynamics become progressively less stable: fixed-point or periodic behavior at larger ν is replaced by stronger sensitivity to perturbations and unstable trajectories at smaller ν, indicating a transition toward higher-dimensional chaotic attractor dynamics.

Time-scale estimation

To estimate a characteristic time-scale of significant separation in non-trivial dynamics cases, the norm of the perturbation $||u_1 - u_2||$ was measured, and the time-scale is identified as the time at which the perturbation grows more than 10% of the mean attractor size. Averaging on several trials, giving the randomness of the perturbation, the results are reported in the table.

Table 2: Perturbation time-scales.

ν	$6.5 \cdot 10^{-4}$	$2 \cdot 10^{-4}$
$t_{\text{perturb}}/\nu\,[s^2 m^{-4}]$	84.1	35.7

$\underline{\nu = 10^{-2}}$:

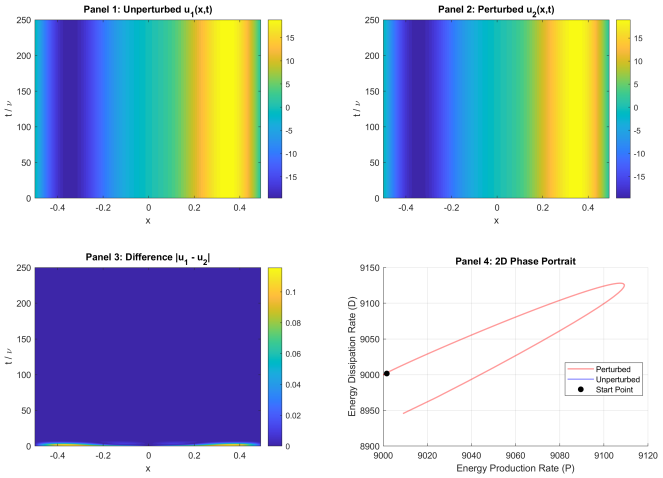


Figure 13: Plot A1

$\underline{\nu = 10^{-3}}$:

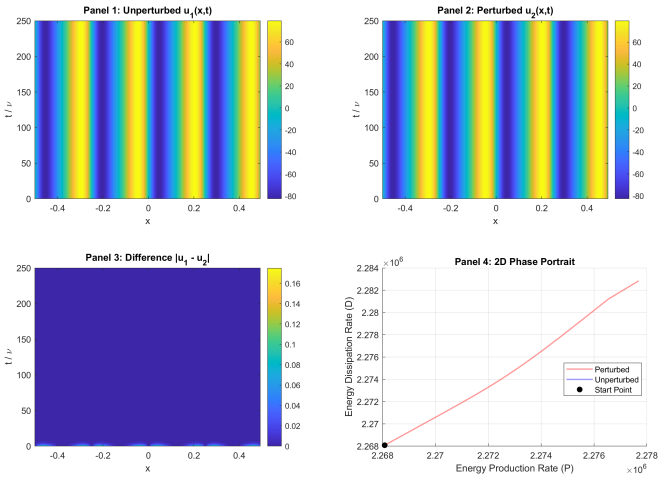


Figure 14: Plot A2

$\underline{\nu = 8 \cdot 10^{-4}}$:

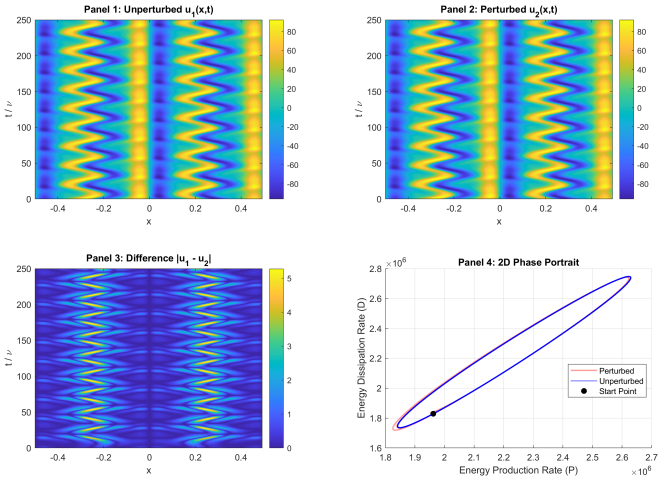


Figure 15: Plot A3

$\underline{\nu = 6.5 \cdot 10^{-4}}$:

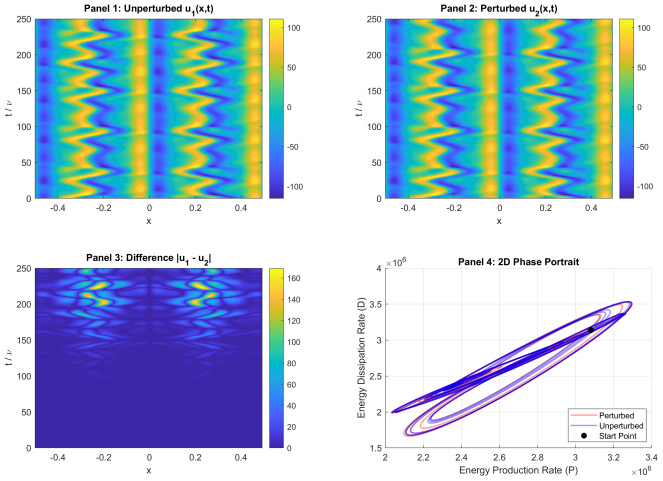


Figure 16: Plot A4

$\underline{\nu = 2 \cdot 10^{-4}}$:

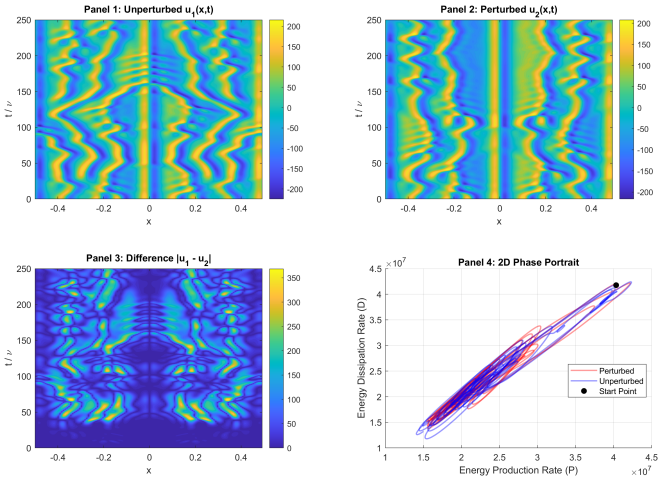


Figure 17: Plot A5

3.1.2 Quantitative characterization

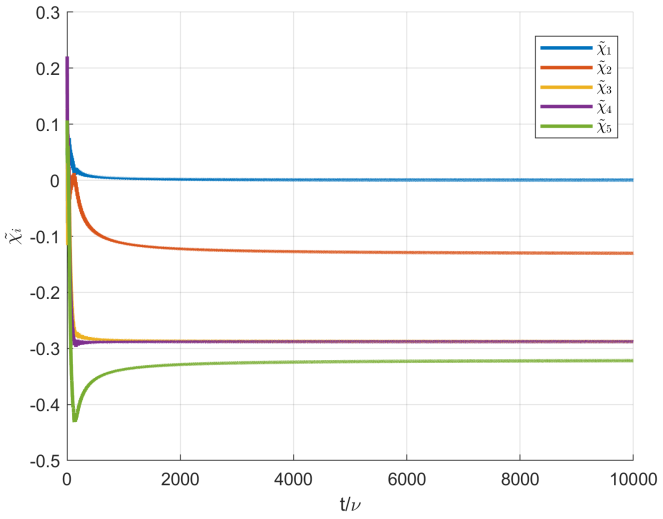


Figure 18: Plot B1: $\nu = 8 \cdot 10^{-4}$

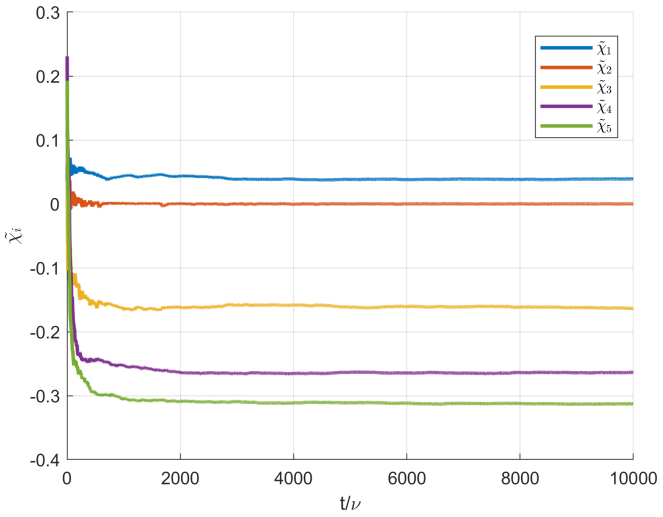


Figure 19: Plot B2: $\nu = 6.5 \cdot 10^{-4}$

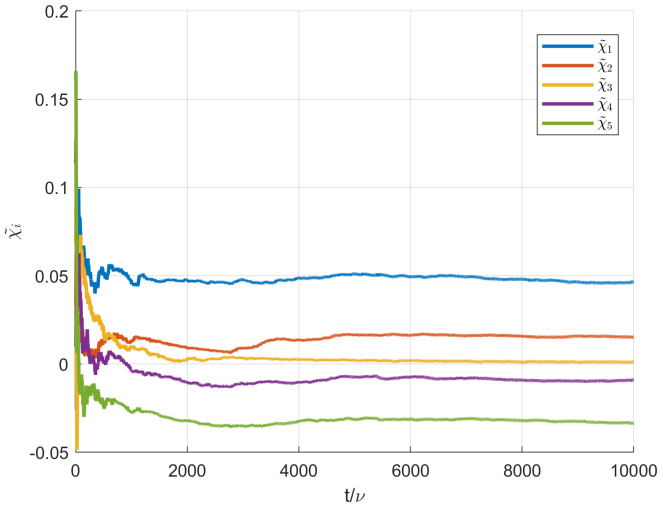


Figure 20: Plot B3: $\nu = 2 \cdot 10^{-4}$

Table 3: The 5 leading rescaled Lyapunov exponents.

ν	$8 \cdot 10^{-4}$	$6.5 \cdot 10^{-4}$	$2 \cdot 10^{-4}$
$\tilde{\chi}_1$	0.000256308	0.0391391	0.0464567
$\tilde{\chi}_2$	-0.130258	0.000144796	0.0153129
$\tilde{\chi}_3$	-0.287691	-0.163637	0.00118077
$\tilde{\chi}_4$	-0.288031	-0.263578	-0.00901695
$\tilde{\chi}_5$	-0.321828	-0.312859	-0.0333351

Discussion 1—The number of zero rescaled Lyapunov exponents: Since there are no continuous spatial symmetries in the KSE system, the only expected zero Lyapunov exponent is the one associated with the flow direction: it represents a perturbation of the form $w_0 = f(u_0)\Delta t$, which shifts the unperturbed trajectory $\Phi_t(u_0)$ forward in time. In the first case, the only Lyapunov exponent close to 0 is the first one, with value 0.00025. Thus, as guessed in previous section, this dynamic could cor-

respond to a limit cycle, with $\tilde{\chi}_1 = 0$, corresponding to the zero Lyapunov exponent, and $\tilde{\chi}_i < 0 \; \forall i < 1$ corresponding to contractive directions. In the second case, the zero Lyapunov exponent is clearly recognizable, since $\tilde{\chi}_1 > 0$ corresponds to the chaotic direction that creates the chaotic dynamic observed before, $\tilde{\chi}_2 \approx 0$ is the zero exponent, and the other values are negative corresponding to contractive directions toward the attractor. Finally, in the third case, that displayed a clearly chaotic dynamic in previous section, the value closer to 0 is the third one: the first two positive exponents are both chaotic directions, the third is the zero exponent, and the last ones contractive directions.

Discussion 2—The maximum rescaled Lyapunov exponent: As ν decreases, the maximum Lyapunov exponent increases, indicating that the exponential rate of growth in the chaotic direction is higher, thus the solution will be more sensible towards perturbations of the initial condition, leading to a more chaotic dynamic, as showed qualitatively also before.

Discussion 3—chaos vs hyperchaos: In the first chaotic regime ($6.5 \cdot 10^{-4}$), I observe one positive exponent, while in the second one ($2 \cdot 10^{-4}$) two. In fact, the difference between chaos and hyperchaos lies in the number of exponential growth directions: in choas we have one positive Lyapunov exponent, i.e. one chaotic direction, while, in the hyperchaos case, we have multiple directions in which perturbations grow exponentially.

Discussion 4—The Lyapunov time: The Lyapunov times for the chaotic cases are reported in the table below. The relevant time-scale found before, for the system with $6.5 \cdot 10^{-4}$, is $t_{\text{perturb}} = 0.0547$ and has the same order of magnitude of t_L^*, $\sim O(10^{-1})$, even if slightly more than it. Considering the second case, $t_{\text{perturb}} = 0.0063$, again, is slightly more than t_L^*, but with the same order of magnitude $O(10^{-2})$. This confirms that the time scales computed before are consistent estimates of perturbation times.

Table 4: Lyapunov times.

ν	$6.5 \cdot 10^{-4}$	$2 \cdot 10^{-4}$
$t_L^* = \frac{1}{\chi_{max}}[s]$	0.0166	0.0043

3.2 Periodic solutions of the KSE

3.2.1 Continuation of the limit cycle

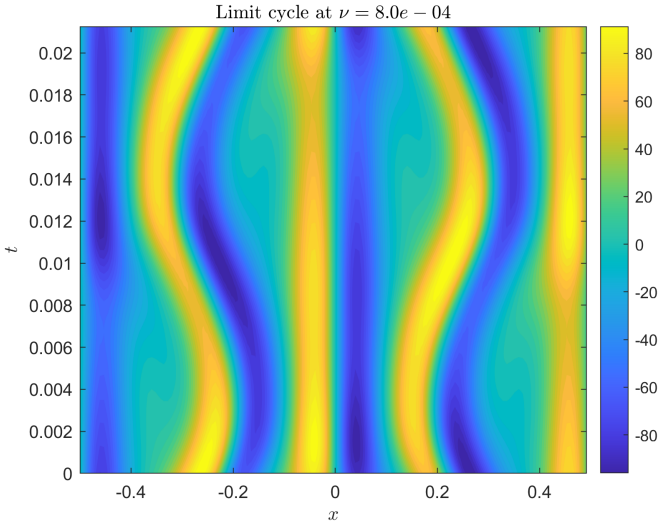


Figure 21: Plot C

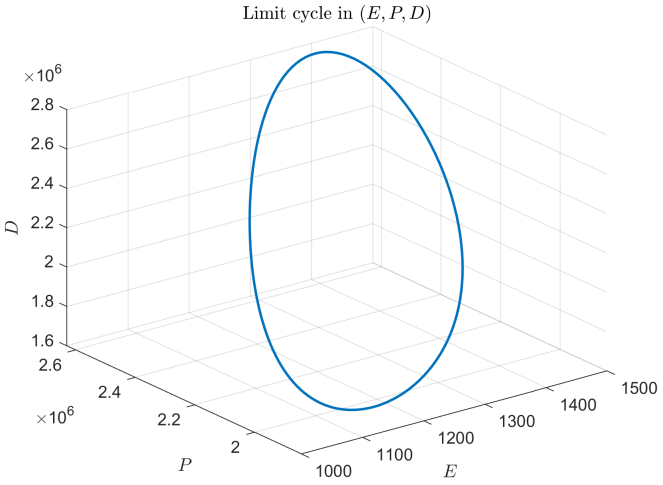


Figure 22: Plot D

ν	T
$8 \cdot 10^{-4}$	$2.125141 \cdot 10^{-2}s$
$7.9 \cdot 10^{-4}$	$2.177307 \cdot 10^{-2}s$
$7.0 \cdot 10^{-4}$	$1.899002 \cdot 10^{-2}s$
$6.5 \cdot 10^{-4}$	$1.604459 \cdot 10^{-2}s$
$6.0 \cdot 10^{-4}$	$1.445473 \cdot 10^{-2}s$

Table 5: Pairs (ν_i, T_i) for $i = 0, 5, 10, 15, 20$

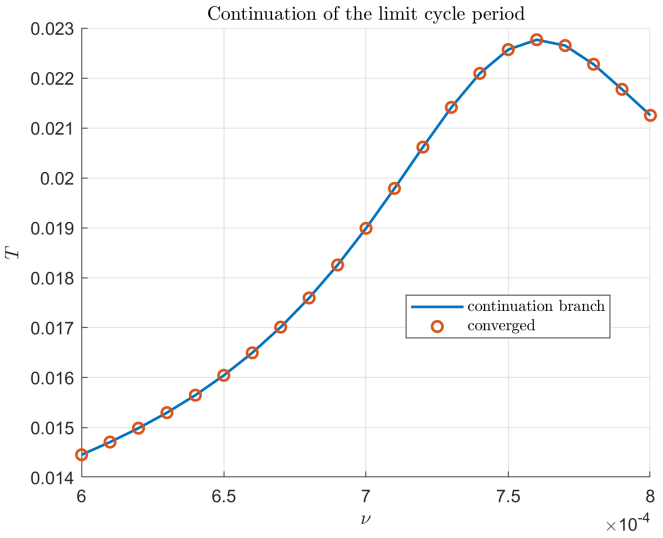


Figure 23: Plot E

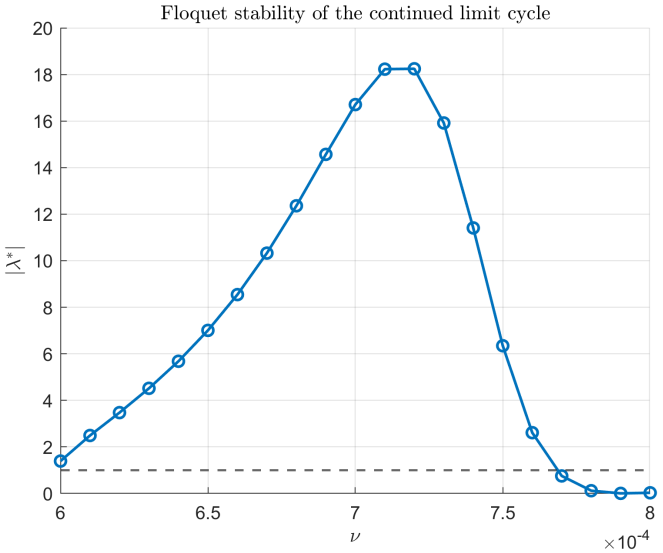


Figure 24: Plot F

Discussion:

- The modulus of λ^* crosses 1 at approximately $\nu = 7.686772 \cdot 10^{-4}$. This means that for $\nu = 8 \cdot 10^{-4}$ the periodic orbit is stable as guessed in section 3.1, as in this case the module of the largest Floquet

ν	$\lvert\lambda^*\rvert$
$8.0 \cdot 10^{-4}$	$2.972869 \cdot 10^{-2}$
$7.5 \cdot 10^{-4}$	$6.346648 \cdot 10^{0}$
$7.0 \cdot 10^{-4}$	$1.670663 \cdot 10^{1}$
$6.5 \cdot 10^{-4}$	$7.006468 \cdot 10^{0}$
$6.0 \cdot 10^{-4}$	$1.393133 \cdot 10^{0}$

Table 6: Pairs $(\nu_i, |\lambda_i^*|)$ for $i = 0, 5, 10, 15, 20$

multiplier is less than 1. For values of the hyperviscosity below the threshold, the periodic orbit becomes unstable: it is no longer a limit cycle, and nearby trajectories separate from it, even if slowly if $|\lambda^*|$ is slightly above 1.

- The passage to chaos is "continuous", meaning that the dynamic goes continuously from a limit cycle, to an unstable orbit, with trajectories that slowly deviate from it, to unstable orbits, onto which separation is really fast.

3.2.2 Periodic orbits

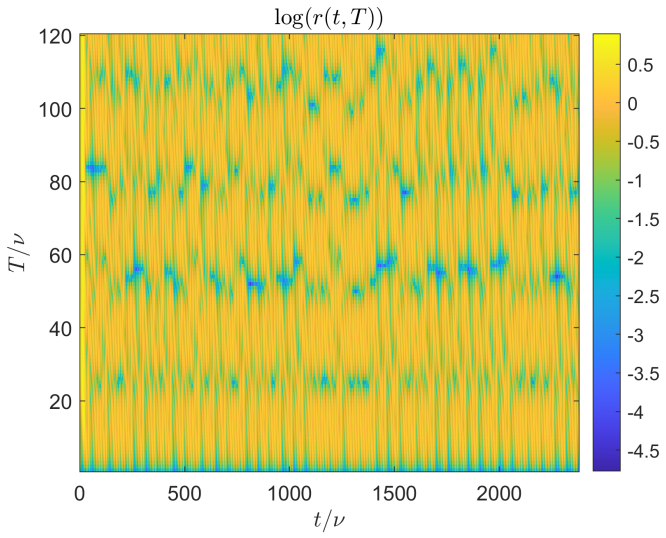


Figure 25: Plot G

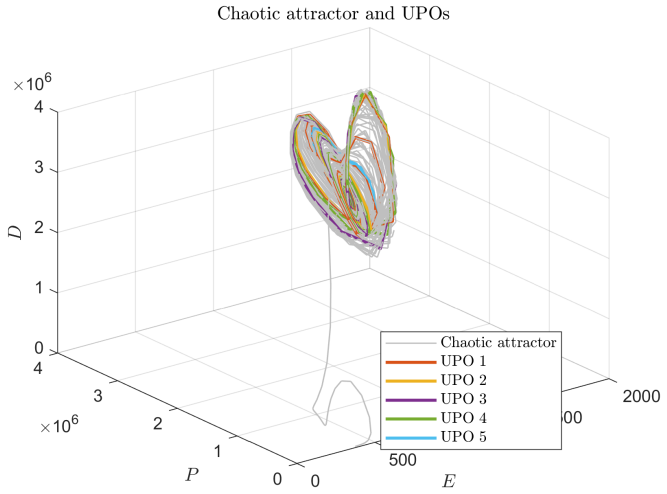


Figure 26: Plot H

t	T_{guess}	search result	T_{UPO}	$\max\{\lvert\lambda_i\rvert\}$	stability	t_F^*
$1.0088 \cdot 10^{0}$	$5.0050 \cdot 10^{-2}$	3	$4.996296 \cdot 10^{-2}$	$9.488196 \cdot 10^{1}$	unstable	$1.097452 \cdot 10^{-2}$
$5.2845 \cdot 10^{-1}$	$3.3800 \cdot 10^{-2}$	3	$3.396562 \cdot 10^{-2}$	$1.176005 \cdot 10^{1}$	unstable	$1.378079 \cdot 10^{-2}$
$9.3275 \cdot 10^{-1}$	$3.7050 \cdot 10^{-2}$	2	$3.736716 \cdot 10^{-2}$	$6.996493 \cdot 10^{0}$	unstable	$1.920787 \cdot 10^{-2}$
$6.2010 \cdot 10^{-1}$	$6.8900 \cdot 10^{-2}$	3	$6.939337 \cdot 10^{-2}$	$2.850736 \cdot 10^{1}$	unstable	$2.071343 \cdot 10^{-2}$
$8.8010 \cdot 10^{-1}$	$1.6250 \cdot 10^{-2}$	3	$1.604459 \cdot 10^{-2}$	$7.006437 \cdot 10^{0}$	unstable	$8.241393 \cdot 10^{-3}$

Table 7: Unstable periodic orbits.

Discussion:

- All the computed orbits are unstable periodic orbits lying in the region of the attractor explored by the trajectory. Therefore, they show that unstable solutions exist within the observed dynamics. However, their existence alone is not sufficient to prove that the KSE dynamics is chaotic. To rigorously establish chaos, one should show sensitive dependence on initial conditions, topological transitivity, and the density of periodic orbits in the chaotic attractor. Hence, the computed UPOs support the chaotic interpretation, but they alone do not constitute a proof of chaos.
- The Lyapunov time is $t_L^* \approx 25.55\nu$. The values of t_F^* are of the same order of magnitude, although they differ from one orbit to another because they depend on both the period and the largest Floquet multiplier of each UPO. It is expected that they are in a similar range, since the first describes the time scale at which a perturbed solution grows with respect the unperturbed one, while the second one describes how fast the trajectory of a solution, perturbed with respect to an orbit trajectory, which could be considered as an unperturbed solution in this context, separates from it.

Appendix

List of Sources

References

[1] P. O'Neill, D. Nicolaides, Damon Honnery, and Julio Soria. Autocorrelation functions and the determination of integral length with reference to experimental and numerical data. 01 2004.

[2] Stephen B. Pope. *Turbulent Flows*. Cambridge University Press, Cambridge, 2000.

List of Collaborators

Stefano Bernasconi, Alex Dell'Orto, Francesco Derme.

Personal Statement

I hereby certify that I fully respect the stated Honor Code and specifically that:

1. My report is my original work prepared solely by me;
2. All sources used are cited;
3. All people I collaborated with are listed.

Signature (Sofia Sannino) Date